



Increasing density of juvenile Atlantic (*Gadus morhua*) and Greenland cod (*G. ogac*) in association with spatial expansion and recovery of eelgrass (*Zostera marina*) in a coastal nursery habitat

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ABSTRACT

Global declines of eelgrass (*Zostera marina*) have major ramifications for the juvenile fishes for which this plant can provide nursery habitat. However, it is less clear how rapidly fishes can recover when habitat loss is reversed, either through natural growth or through habitat enhancement efforts. We investigated the consequences of natural expansion and recovery of eelgrass following disturbance, on the densities of juveniles of two cod species – Atlantic (*Gadus morhua*) and Greenland (*Gadus ogac*) cod – in a coastal nursery area in Newfoundland, Canada. Eelgrass has been expanding at this location, likely as a result of reduced disturbance from winter pack-ice in the past decade. We conducted bi-weekly sampling of juvenile fish densities at 12 sites from July to November annually, 1995–2006, by seine netting. Each site was designated as either 'natural', 'expansion', 'removal-recovery' or 'unvegetated' dependent on their recent eelgrass coverage and involvement in a manipulation study (1999–2000; Laurel et al. 2003b). We measured eelgrass percentage cover at each site by analysis of aerial photographs. Eelgrass cover had noticeably increased in the 1999–2000 'removal-recovery' sites and at the 'expansion' sites following 2000. Using randomized intervention analysis (RIA) we detected positive relationships between eelgrass cover and relative fish density during 2001–2006. These results suggested a significant and rapid increase in age-0 cod density associated with eelgrass habitat expansion. We suggest that juvenile cod density responds to annual expansion and contraction of eelgrass cover, illustrating the potential cascading effects of eelgrass loss associated with climate change, human disturbance, and disease and the potential benefits of eelgrass conservation and enhancement.

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1. Introduction

Seagrass beds are well known as productive habitat components in coastal marine ecosystems (den Hartog, 1970; Kikuchi, 1980; Fonseca et al., 1982). The elevated productivity and associated functions of water clarification, sediment stabilization and reduced water flow are provided by extensive rhizome and root system development below the sediment-water interface and by the emergent blades that provide shelter, primary production, and food resources above this interface. Eelgrass – *Zostera marina* – is the most widespread (McRoy and Helfferich, 1977) of numerous seagrass species, and spans north temperate latitudes in both the North Pacific and North Atlantic oceans as far north as the Arctic Circle in some areas of Europe (den Hartog,

1970). Eelgrass grows in shallow subtidal and intertidal areas at depths typically shallower than 5–10 meters (den Hartog, 1970; Phillips and Meñez, 1988). The nearshore location of eelgrass beds exposes this habitat to frequent disturbances from wave and tide action (Fonseca et al., 1983), as well as pack-ice in some areas of its range.

Seagrasses are declining globally (Short and Neckles, 1999; Hemminga and Duarte, 2000; Duarte, 2002; Short et al., 2006). The large-scale decline in eelgrass meadows throughout the North Atlantic in the 1920s and 1930s has been attributed to "wasting disease" (den Hartog, 1987; Short et al., 1987; Muehlstein et al., 1991) and a pathogenic strain of *Labyrinthula* that remains ubiquitous in eelgrass meadows worldwide (Short et al., 1987). Direct anthropogenic influences have also contributed to the decline in seagrass beds, through eutrophication, introduced species, climate change, shoreline development, dredging, and other physical disturbances (Orth and Moore, 1983; Dennison et al., 1993; Duarte, 1995; Nixon, 1995; Short and Wyllie-Echeverria, 1996; Hemminga and Duarte, 2000). The loss of eelgrass also destabilizes shorelines (Cardoso et al., 2004) and has

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been linked to sudden and substantial declines in fish densities in formerly productive nursery habitats in Europe (Muehlstein, 1989).

In eastern Canadian coastal waters, the current state of eelgrass parallels the global trend. Declines in eelgrass have been noted throughout its range except near its northerly limits, such as the island of Newfoundland (Hanson, 2004). In Newfoundland waters eelgrass habitats function as important nurseries for juvenile Atlantic cod (*Gadus morhua*) following their settlement to demersal habitat after a pelagic larval and early juvenile phase (Gotceitas et al., 1997; Laurel et al., 2003b; Laurel et al., 2004). Predation risk decreases (Gotceitas et al., 1995; Linehan et al., 2001; Laurel et al., 2003b) and food availability increases locally (Levin et al., 1997) in eelgrass habitats.

Juvenile age-0 Atlantic cod typically occur in coastal Newfoundland waters from July to December (Methven and Bajdik, 1994; Gotceitas et al., 1997; Grant and Brown, 1998; Gregory et al., 2006), and appear in nearshore Newfoundland habitats in one or more settlement pulses throughout this period (Laurel et al., 2003b; Ings et al., 2008). Greenland cod (*Gadus macrocephalus ogac*) – recently identified as a subspecies of Pacific cod, *Gadus macrocephalus* (Coulson et al. 2006) – is a northern species that in the Atlantic Ocean inhabits shallow coastal waters (Scott and Scott, 1988) from the Canadian arctic and Greenland to Cape Breton, Nova Scotia. Larvae are pelagic and then settle into shallow seabed habitats in a single pulse (Laurel et al., 2003b, 2004; Ings et al., 2008). Demersal age-0 juveniles of both species occupy eelgrass nursery habitats significantly more than alternative nearby habitats (Gotceitas et al., 1997). Increases in eelgrass habitat may be reasonably expected to increase survival of vulnerable, post-settled juveniles during their residency in shallow, coastal marine habitats. The loss of eelgrass habitat has potential negative consequences on juvenile habitat productivity for the young of numerous fish species, including cod.

Restoration or even “creation” of eelgrass habitat has obvious appeal to habitat managers and developers, and there has been significant effort to recover disrupted seagrass habitat around the world (Davis and Short, 1997; Fonseca et al., 1998; Short et al., 2002). However, it is unclear how rapidly fish populations can respond to reversal of habitat loss through natural or fostered regrowth or through other habitat enhancement efforts.

In our study, we analyzed an 11-year coastal fish survey dataset – 1995–2006 – to investigate the consequences of naturally increasing eelgrass beds on densities of juvenile cod. Such a large data set poses issues of temporal variability and so it is necessary to treat the highly variable data with careful consideration during analysis. We also took advantage of an eelgrass habitat removal and enhancement experiment conducted during 1999 and 2000 in a previous study (Laurel et al., 2003a,b; Laurel et al., 2004) at four of our sites – to investigate the potential for increase of juvenile fish densities with habitat recovery following the simulated disruption of eelgrass habitat via the removal of above-ground biomass. We hypothesize that abundance of juvenile fishes in recovering or expanding eelgrass sites increase as eelgrass cover increases, relative to unmanipulated eelgrass and unvegetated sites.

2. Methods

Newman Sound is a ~41 km long, 1–3 km wide fjord in Bonavista Bay, located on the northeast coast of the island of Newfoundland, Canada (Fig. 1). The sound is composed of an inner and outer basin characterized by maximum depths of 55 and 300 m, respectively, which are separated by an 18-m deep sill. The dominant nearshore subtidal substrate in the inner sound is mud, sand, and gravel, interspersed with broad expanses of exposed bedrock. In our study area, eelgrass grows subtidally to a maximum depth of ~6 m along 40% of this shoreline mainly in the south and west (Fig. 1). Almost the entire north shore in the outer sound consists of rocky substrate unsuitable for eelgrass growth. Mud and mud-gravel suitable for

eelgrass growth occupies only ~5% of the shore in the shallow subtidal zone of the outer sound and this vegetation is restricted to the heads of a few coves along the southern shore.

Our study encompassed 12 sites; seven in the inner and five in the outer sound (Fig. 1). Six sites were undisturbed “eelgrass” sites (55–91% seabed coverage). Three sites were “expansion” sites – initially barren of vegetation at the beginning of our study (0% coverage in the late 1990s) – within which eelgrass growth occurred during the study period (up to 83% coverage during or before 2006). Although one of the three sites (MS) did not have as much eelgrass as the other two sites throughout our study period, by 2008 eelgrass had increased by an order of magnitude to >30% coverage over that of two years previous and so we have designated it an “expansion” site. One site remained “unvegetated” throughout our study (i.e., 0% eelgrass coverage). The study included two “recovery” sites from which an area of ~500 m² eelgrass was completely removed in 1999 and 2000 by Laurel et al. (2003a,b), but had since regrown from 43% up to 94% coverage by 2006. Two of the three expansion sites that were initially devoid of appreciable vegetation were enhanced with artificial eelgrass (~275 m² area) in 1999 and 2000 (Laurel et al., 2003b). Laurel et al. (2003b) had removed the artificial eelgrass immediately following the completion of their study in autumn 2000.

We collected aerial photographs over each of our sites in six of our study years – 1998–2002, and 2006 – from either a DeHavilland Beaver or a Cessna sea plane at an altitude of 213 m (± 3 m). Photos were nominally taken during low tide periods (near the dates of new or full moons), close to mid-day to maximize incident light level, minimize surface glare and avoid the afternoon winds common to this area of the coast. We used a 35 mm Pentax FX10 camera with a 28–80 mm Pentax F-200 N zoom lens (set at 50 mm). The zoom lens was fitted with haze and polarizing filters to reduce surface glare. Photographs were taken looking directly downward except in 1998–2000 in which images were captured obliquely and ground markers used to facilitate photo image processing. Each site was photographed while flying into the wind in order to reduce speed and wind turbulence and thus increase image clarity.

We determined eelgrass cover in each site using the “box counting” method (Scheuring and Riedi, 1994), in which a grid with squares measuring 2 by 2 m (4 m²) each were placed over a 1250 m² sampling area in each digital image. The presence or absence of eelgrass in each square was recorded. In 1998–2001, three ground markers painted with fluorescent paint were deployed at high tide prior to collection of aerial photographs. At each site, two markers were positioned 25 m apart on shore at the high tide mark. A third marker was positioned 50 m offshore at the surface at right angles to one of the shore markers, marking an area of 1250 m². This approach permitted any distortion to be corrected during image analysis (especially with oblique photographs). We used Adobe Photoshop CS2™ to manipulate and analyze photographs.

Annual percentages of eelgrass cover for 2003–05 were estimated from simple linear interpolation. Image quality in the 1998 and 1999 photographs was generally poor for most of our sites. The years 1998 and 1999 were a period of relatively stable eelgrass habitat configurations in our study area. Therefore, we hindcasted our estimates of eelgrass percent cover from Sheppard (2002), who analyzed aerial photographs taken at our sites during 1999–2000, and groundtruthed from visual surveys by boat of percent eelgrass coverage in 1995–96 (Gregory et al., 1997).

Densities of age-0 juvenile cod were estimated from tows of a 25-m demersal seine net deployed from a small boat (5–6 m). The net was set ~55 m from the shore with two people located 16 m apart on the beach. The seine was set so that it would sample the bottom 2 m of the water column, thereby including all eelgrass patches in its path. A total area of 880 m² was sampled for each set using this method; capture efficiency for juvenile cod was 95% using this method (Gotceitas et al., 1997; Laurel et al., 2003b). Captured fish were quickly removed from

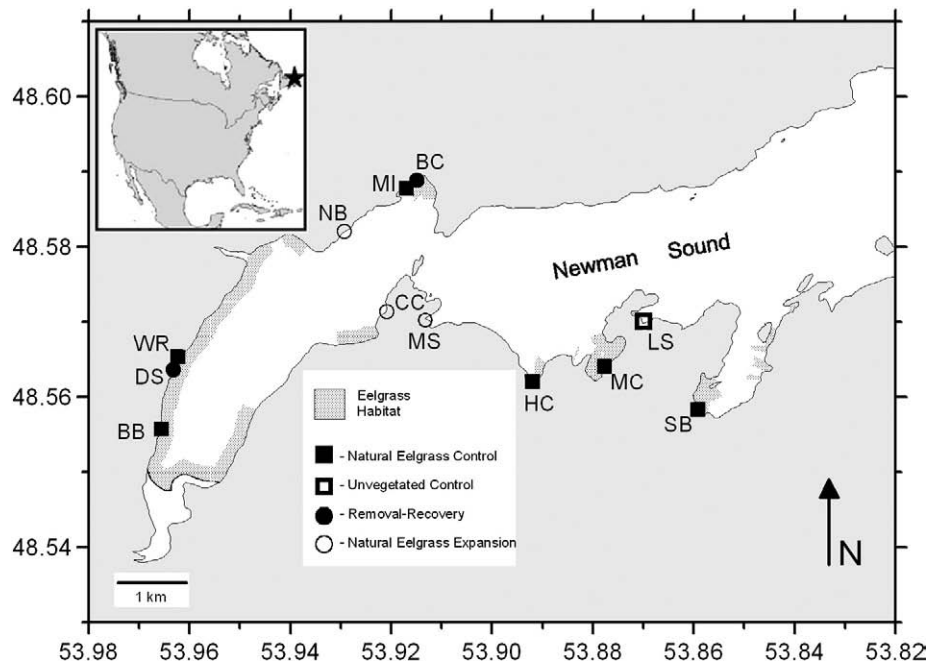


Fig. 1. Study area in Newman Sound, Bonaville Bay, Newfoundland showing the locations of twelve (12) sites sampled, 1995–2006 (Big Brook (BB), Buckley's Cove (BC), Cannings Cove (CC), Dockside (DS), Hefferns Cove (HC), Little South Broad Cove (LSB), Mistaken Cove (MI), Minchin Cove (MC), Mount Stamford (MS), New Bridge Cove (NB), South Broad Cove (SB), and White Rock (WR); DS/BC were removal and CC/MS were eelgrass addition sites during 1999–2000 (see [Laurel et al. 2003b, 2004](#)).

the net and transferred to a fish tub filled with clean seawater. Each fish was quickly identified, measured, and placed into a second seawater tub prior to subsequent release. Seine samples were collected bi-weekly from mid-July to late November each year 1995–2006 (no sampling in 1997), providing a nominal total of 10 collections at each site every year (1320 total fish collections in 11 years). We evaluated changes in the density of juvenile cod associated with recovery from the 1999–00 habitat manipulation and natural expansion of eelgrass beds in our sites in the following six year period, 2001–06. The area of removal was 500 m², the area of addition was ~275 m², which in appropriate sites were both nested within a seined area of 880 m² to maintain the protocol used in all previous years at these sites. The seined area, in turn, was nested within a 25 × 50 m area (1250 m²), in which eelgrass percentage cover was determined from aerial photographs. Given this technique, eelgrass cover for the removal years was ~43%, not 0%.

We used Randomized Intervention Analysis ([Carpenter et al., 1989](#)) adhering to the principle of a Before-After-Control-Impact (BACI) design ([Stewart-Oaten et al., 1986](#)) to analyze annual trends in the catch data. All juvenile fish density data were log transformed [$y = \log_{10}(x + 1)$] before analysis because of high inter-annual and inter-site variance. Differences between fish abundances at the experimental (i.e., recovery and expansion sites) and reference eelgrass sites were determined for each year of the study. The mean values for these differences were then compared between pre-manipulation ($\bar{x}_{(pre)}$) and post-manipulation ($\bar{x}_{(post)}$) years. A test statistic was generated for each comparison, $|\bar{x}_{(pre)} - \bar{x}_{(post)}|$, and compared to 5000 random permutations of the experimental-reference site data to obtain a *p*-value. RIA tests were used for the following combinations of years, 2001–2002, 2003–2004, and 2005–2006.

We also determined relative differences (RD) in fish density between experimental (recovery and expansion sites) and reference (control) sites (eelgrass and unvegetated sites). RD values can be interpreted as order of magnitude level effects considering that the data were log transformed prior to analysis. For example, RD = 1.0 would mean that juvenile fish density was an order of magnitude (i.e., 10x) higher than the corresponding reference site. An RD of zero

would indicate similarity between the experimental and reference sites. RIA derives its error distribution from a set of randomizations of the collected data, rather than on residuals; therefore, non-normality does not affect statistical test results ([Edgington, 1980](#); [Carpenter et al., 1989](#)).

3. Results

3.1. Interannual changes in eelgrass cover

Percent cover at the six natural eelgrass sites exceeded 55% ([Fig. 2](#)), and generally remained stable (S.D. ~5% among years) throughout most of the 11-year period (no data in 1997). One barren site – LSB – had no eelgrass at any time during the study period. The first occurrence of eelgrass differed among the three expansion sites where no eelgrass had been present in the previous years in our databases – CC in 2000, MS in 2002, and NB in 2004. In all three expansion sites, eelgrass was first observed during annual site visits (e.g., [Gregory et al., 2006](#)), and eelgrass aerial extent increased following first occurrence (e.g., from zero to 78% in 2 years at CC; from zero to 83% in 3 years at NB – [Fig. 2](#)). Eelgrass percent cover for the two removal sites – DS and BC – could not be hindcasted to include 1998–1999 because those years immediately preceded the manipulation years (1999–2000, see [Laurel et al., 2003b](#)). Removal sites showed a rapid recovery of eelgrass coverage, probably due to the technique used by [Laurel et al. \(2003b\)](#), which intentionally left the eelgrass rhizomes intact at these sites.

3.2. Juvenile fish density

Density of age-0 cod among all sites was highly variable (from 0 to 1300 fish/880 m²) among the 11 years of our study ([Fig. 3](#)). Higher densities of cod were observed in both 1999 and 2002, in contrast with other years in the dataset. Interannual variation in density between Atlantic cod and Greenland cod was similar across years independent of eelgrass cover, suggesting that this variation in juvenile density was driven largely by environmental factors common to both species ([Fig. 3](#)).

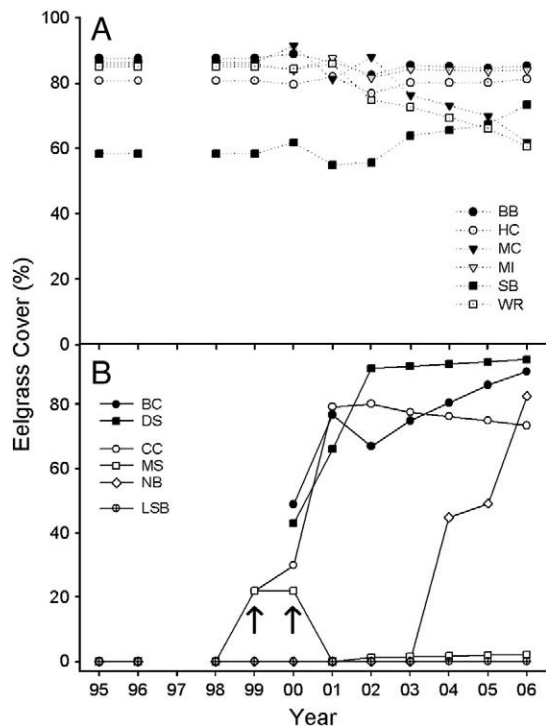


Fig. 2. Annual percent eelgrass cover at A natural eelgrass and B removal–recovery (closed symbols), expansion (open symbols), and unvegetated sites (cross hatched symbol) over an 11 year period. (No data was available for 1997; 1995–96 and 1998–99 data were hindcasted using Sheppard (2002) and Gregory et al. (1997); 2000–02 and 2006 data were obtained from aerial photographs; 2003–2005 data were interpolated between photograph years). Arrows indicate Laurel et al. (2003b) manipulation years (removal and addition).

3.3. Eelgrass recovery

Age-0 cod densities increased in recovery sites – BC and DS – soon after 1999–2000 and then stabilized. Significant decreases in fish abundance from 1998 to 1999 and 2000 reported by Laurel et al. (2003b) in both sites were evident in our recalculations for these sites

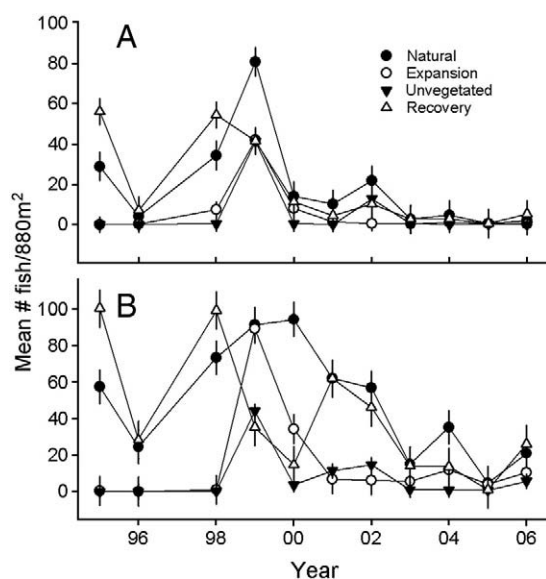


Fig. 3. Interannual mean density ($\pm$ SE) of A *G. morhua* and B *G. ogac* at each of four conditions of vegetation in Newman Sound, Newfoundland 1995–2006. Natural eelgrass ($n=6$), eelgrass expansion ($n=3$), unvegetated ($n=1$), and recovery sites ($n=2$). Refer to Fig. 1 for site names.

as well (Fig. 4). RD values for both sites increased in the post-manipulation years toward pre-manipulation levels (Fig. 4), in two of four cases significantly (*G. morhua*, BC $p<0.05$; *G. ogac*, DS $p<0.01$). Greenland cod appear to recover more quickly after eelgrass removal than do the Atlantic cod.

3.4. Eelgrass expansion

RD values at the three expansion sites increased steadily following the first observation of eelgrass at these sites (2000–CC, 2002–MS, and 2004–NB), compared with control sites in most cases significantly (Fig. 5; *G. morhua*, NB $p<0.01$, MS $p<0.05$; *G. ogac*, NB $p<0.001$, CC $p<0.001$, MS $p<0.01$). For the earliest years (1995–1998), annual mean RD values were typically near -1.0 . By 2006 these values had generally increased by one order of magnitude to zero for both Atlantic and Greenland cod juveniles – i.e., a ten-fold density increase above the pre-expansion years. In the final year of the study, these sites had densities of juvenile cod similar to those at the eelgrass control sites.

Laurel et al. (2003b) reported significant increases in fish densities at the two “enhancement sites” in the two years – 1999 and 2000 – when artificial eelgrass patches were added to “non-vegetated” sites. Artificial eelgrass was removed at the conclusion of their experiment in 2000. As expected, we observed a substantive decline in fish density at these two sites – CC and MS – in 2001 (the following year) consistent with the pre-manipulation condition (Figs. 5 and 6). It also appears that Greenland cod respond more quickly than the Atlantic cod to this change. As eelgrass expanded in these sites throughout subsequent years, densities of juvenile cod of both species progressively became more similar to “eelgrass” sites than they were to unvegetated sites (Fig. 5).

3.5. Juvenile fish density and eelgrass cover

Among recovery and expansion eelgrass sites, juvenile cod densities increased as the percentage of eelgrass increased, irrespective of the original vegetative condition of the site (Fig. 7). At the two recovery sites (DS and BC), relative difference values increased during the post-removal years (2001–2006; Fig. 4), becoming less different from the pre-removal condition and coinciding with increasing percent eelgrass cover for those years ($\sim 45\%$ to 90% ; Fig. 2). The expansion sites showed a similar pattern, with significant increases in relative cod density during that same period (Fig. 5), when eelgrass was naturally expanding from 0% to $\sim 80\%$ in some cases (CC and NB; Fig. 2). Regardless of the initial vegetative state or location of these sites, juvenile cod density and eelgrass cover covaried. Greenland cod densities tended to be more significantly associated with annual and inter-site differences in eelgrass cover than Atlantic cod (Figs. 3–7).

4. Discussion

Seagrasses are currently in decline in the Mediterranean, Caribbean, Australia and much of the North Atlantic (Duarte, 1999). In contrast to this general global trend, eelgrass habitat in Newfoundland coastal waters through the study period appeared to be expanding, possibly because of warmer mean summertime coastal sea temperatures and reduced winter sea ice conditions that may be associated with climate change (Phillips and Meñez, 1988). As predicted, we observed increases in age-0 cod density in sites where eelgrass habitat was allowed to recover from experimental removal and at sites where eelgrass was expanding naturally. Our results build upon findings by Laurel et al. (2003b), who documented that age-0 juvenile Atlantic and Greenland cod both decline with loss of eelgrass habitat.

Our examination of the two to four years following the recovery and regrowth of eelgrass in these sites showed a significant increase in the percent coverage of eelgrass throughout the study period to

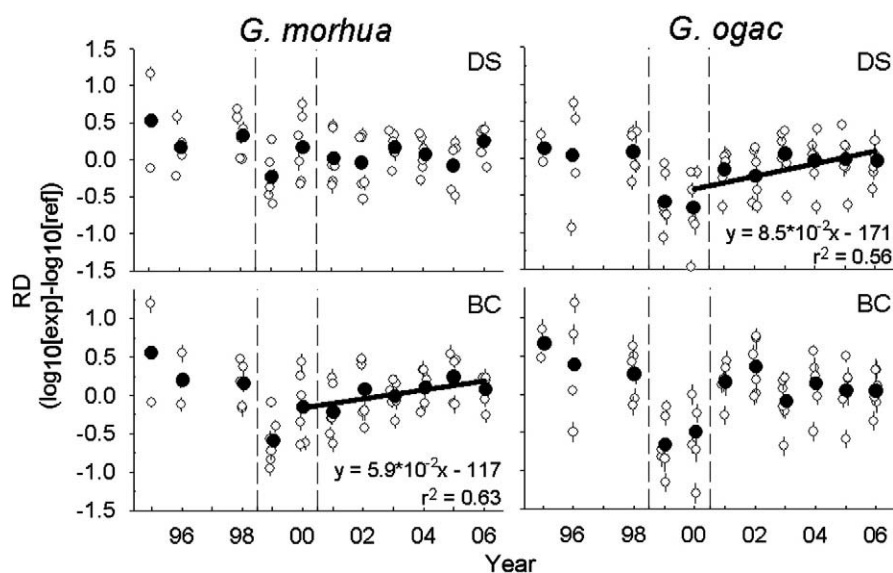


Fig. 4. Relative differences (RD) of age-0 abundances at eelgrass removal–recovery sites (DS and BC) and natural eelgrass reference sites during pre- and post-manipulation years following eelgrass removals in 1999–00 (enclosed by dashed lines) by Laurel et al. (2003b). Open circles represent differences between mean catches ($n = 4–13$) $\pm$ SE each year. Annual means are indicated by closed circles. Points are jittered along the x-axis for visual purposes. Line fits represent significant trends in the abundance of juvenile cod at the experimental site vs reference sites during years of eelgrass recovery (2000–2006).

pre-manipulation levels. Juvenile cod density generally increased with the amount of eelgrass cover. However, the increase occurred over a protracted period (4–6 yrs) suggesting it was a sustained change.

Sites with no eelgrass early in the study period, but into which eelgrass expanded, were occupied by age-0 juvenile cod in proportion to the percent of eelgrass cover. These data suggest that age-0 cod are

generally available to settle in broad regions, including those devoid of eelgrass, but that carrying capacity is likely higher in eelgrass than in unvegetated habitats, limiting site-specific juvenile cod density.

The consequences of habitat limitation on survival and eventual recruitment of juveniles to the adult population remains uncertain, but several studies posit that mortality due to predators is reduced in eelgrass habitat. Over small spatial and temporal scales, seagrass is

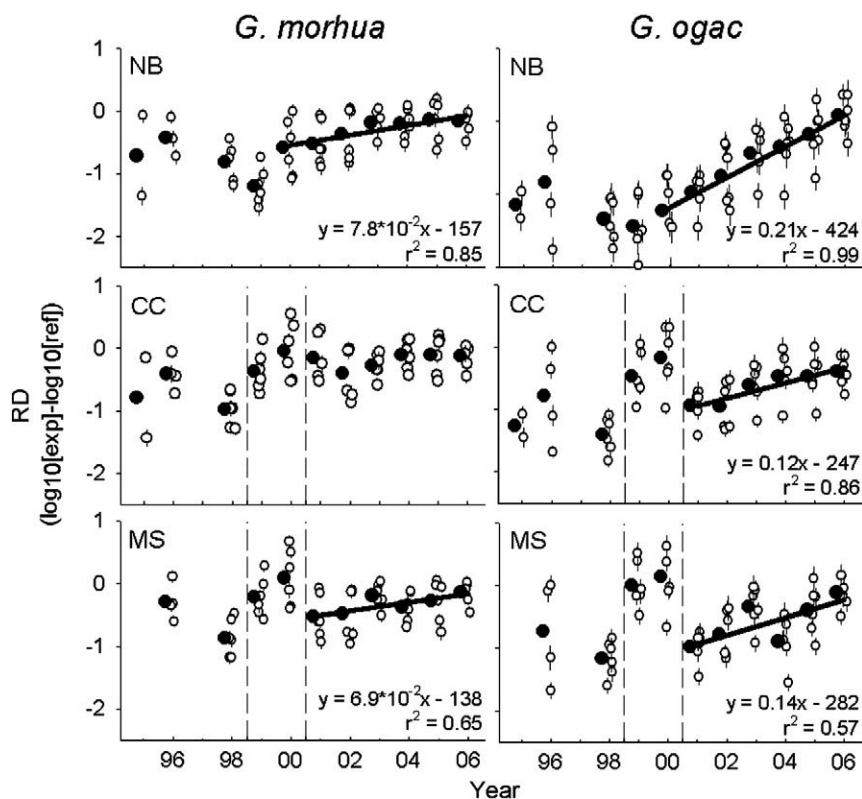


Fig. 5. Relative differences (RD) of age-0 abundances at eelgrass expansion sites (NB, CC, and MS) and natural eelgrass reference sites. The area enclosed by dashed lines indicates years of artificial eelgrass addition (CC and MS; Laurel et al. 2003b). Open circles represent differences between mean catches ($n = 4–13$) $\pm$ SE each year. Annual means are indicated by closed circles. Points are jittered along the x-axis for visual purposes. Line fits represent significant trends in the abundance of juvenile cod at the experimental site vs reference sites during years of eelgrass expansion (2001–2006).

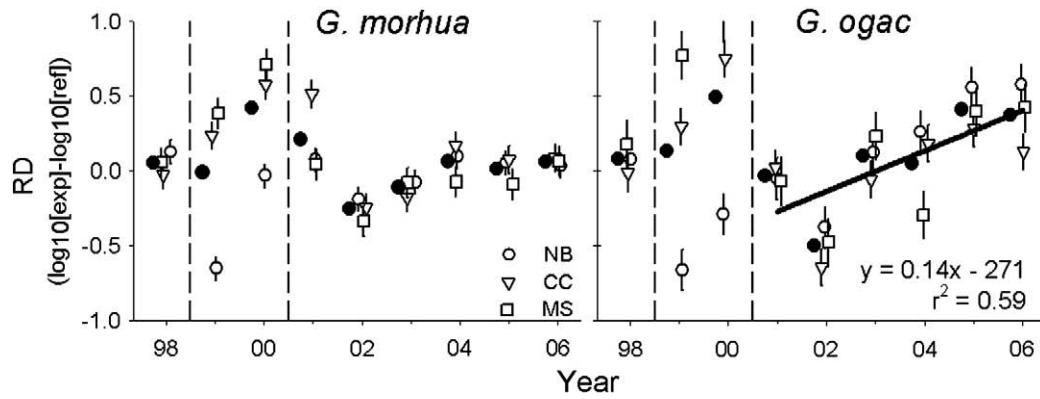


Fig. 6. Relative differences (RD) of age-0 abundances at natural eelgrass expansion sites (NB, CC, and MS), and an unvegetated reference site (LSB). The area enclosed by dashed lines indicates years of artificial eelgrass addition. Individual points represent differences between mean seine catches ($n = 4-13$) $\pm$ SE each year. Annual means are indicated by closed circles. Points are jittered along the x-axis for visual purposes. Line fits represent significant trends in the abundance of juvenile cod at the experimental site vs reference sites during years of eelgrass expansion (2001–2006).

well known to provide protection from predators and entrain invertebrate prey (Gotceitas et al., 1997; Levin et al., 1997; Linehan et al., 2001). Therefore, *a priori* we expected that densities of juvenile cod would be directly linked to eelgrass distribution. However, at larger scales, such links may be relatively inconsequential to the population if egg and larval sources are not in close proximity to these eelgrass nurseries (Bradbury et al., 2008). The population-level significance of our results must be considered in the broader context of local oceanographic conditions and the location of remaining adult cod spawning groups. Methven and Bajdik (1994) identify several possible environmental factors causing variation in near shore juvenile cod abundance between years. These may include wind

speed and direction, hydrographic conditions, and water temperature. It is necessary to control for these effects, although some are beyond the scope of this study. The transformation of our catch data and the use of RIA help to control for some inter-annual variation.

Interspecific differences, while not explicitly addressed in our study, were nevertheless evident. As adults, Greenland cod are associated with shallow coastal waters more so than Atlantic cod (Scott and Scott, 1988). Recruitment dynamics of Greenland cod may be such that habitat associations are more tightly bound to nearshore habitat features than Atlantic cod. Investigating such a prospect was beyond the scope of our study.

Changes in fish density were evident immediately following the termination of eelgrass manipulations conducted by Laurel et al. (2003b) in 2000. Fish density increased in the removal–recovery sites more rapidly than in expansion sites where increases were more protracted in time. It is probable that fish abundance at recovery sites was supported by established eelgrass surrounding the removal area. However, this should be expected. Fish abundance at recovery sites is supported by the well-established eelgrass habitat surrounding the removal area. Natural eelgrass began to grow, or became clearly evident, in two of the enhancement sites (CC in 2000, and MS in 2002) and in one control site (NB in 2004). We observed increased juvenile cod density at these sites eventually attaining levels similar to those at control eelgrass sites. Sensitivity of fish abundance to the extent of eelgrass coverage in our study adds more evidence of the importance of these habitats for settling juvenile cod. Other habitats such as cobble (Tupper and Boutilier, 1995), macroalgae (e.g., Laminaria, Cote et al., 2001) and deep-water emergent structures (e.g., Lindholm et al., 1999) have been suggested to be important for juvenile Atlantic cod survival. However, on the northeast coast of Newfoundland, age-0 Atlantic and Greenland cod are largely restricted to coastal regions where eelgrass is available (e.g., Gotceitas et al., 1995; Laurel et al., 2003b; Linehan et al., 2001). Our study reinforces this link at the fjord scale (~10 km), which is believed to approximate the relevant scale of recruitment in inshore populations (Bradbury et al., 2008).

Our results expand on previous work that has demonstrated the limiting role that eelgrass plays in juvenile cod survival and that its availability potentially influences the recruitment of ecologically and economically important species. Within the context of the current global decline of eelgrass through multiple human-related activities (coastal eutrophication to climate change), this study has shown the importance of preserving or enhancing naturally-occurring eelgrass ecosystems. Our results also demonstrate that relatively rapid recovery of eelgrass and associated juvenile fishes are possible following natural and anthropogenic disturbances in some instances. Although the scales of disturbance and size and distribution of

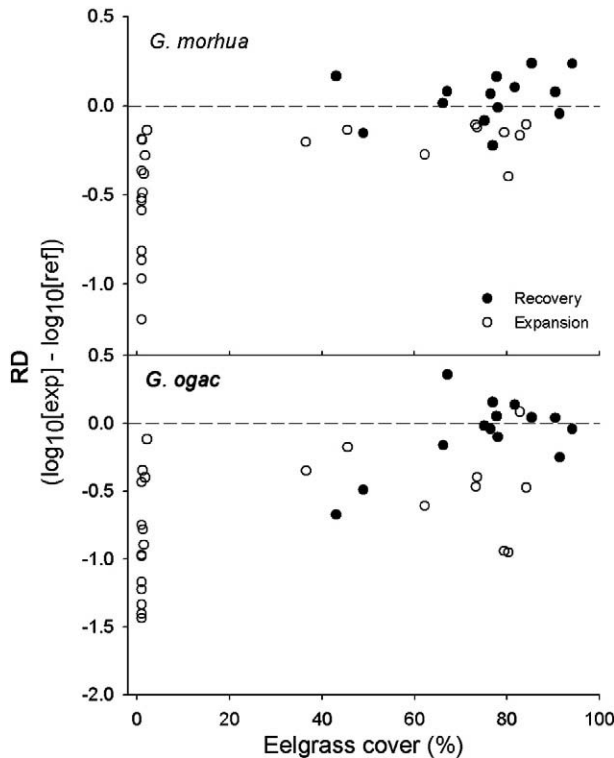


Fig. 7. Relative differences (RD) of age-0 abundances and annual eelgrass percent cover for eelgrass removal–recovery sites and expansion sites across all years 1995–2006. Low eelgrass cover ~10% includes years prior to any expansion or recovery at indicated sites. The dashed line represents zero RD, or no difference from the natural eelgrass sites.

remaining fish populations and other environmental factors almost certainly play significant roles in recruitment to healthy cod populations, the role of healthy habitat components is likely similarly important.

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